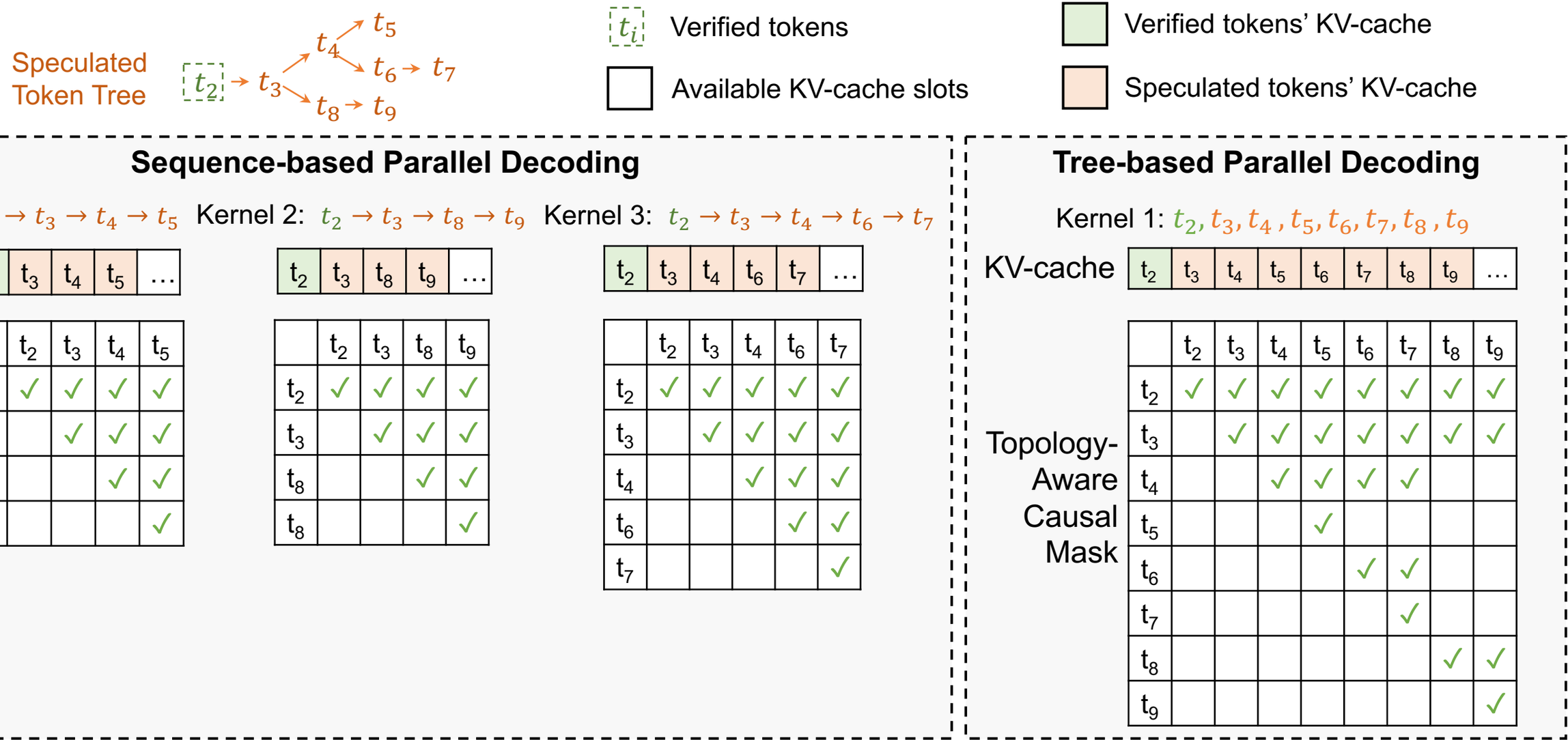
Speculated Token Tree
t_2 → t_3 → t_4 → t_5
t_4 → t_6 → t_7
t_3 → t_8 → t_9
t_i Verified tokens
Available KV-cache slots
Verified tokens' KV-cache
Speculated tokens' KV-cache
Sequence-based Parallel Decoding
Kernel 1: $t_2 \rightarrow t_3 \rightarrow t_4 \rightarrow t_5$
Kernel 2: $t_2 \rightarrow t_3 \rightarrow t_8 \rightarrow t_9$
Kernel 3: $t_2 \rightarrow t_3 \rightarrow t_4 \rightarrow t_6 \rightarrow t_7$
KV-cache
Causal Mask
Tree-based Parallel Decoding
Kernel 1: $t_2, t_3, t_4, t_5, t_6, t_7, t_8, t_9$
KV-cache
Topology-Aware Causal Mask